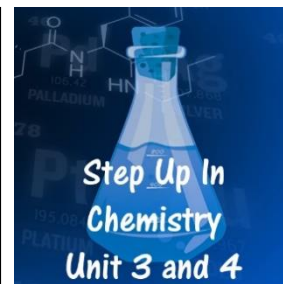


3.3 Galvanic Cells

Problems Worksheet



1. State the role of each of the following in a galvanic cell:

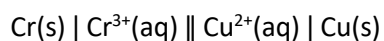
a. Anode

b. Cathode

c. Salt bridge

d. External circuit

2. Shorthand notation is often used to describe galvanic cells. Draw a diagram of this cell:



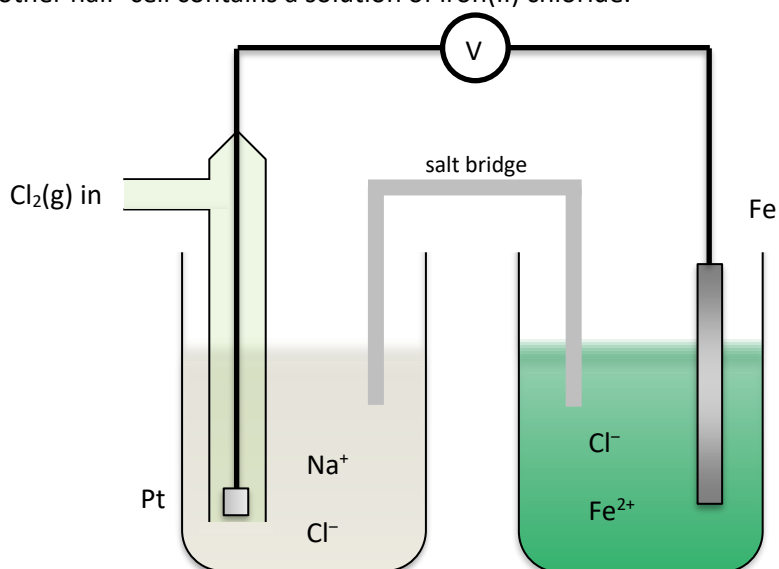
Include the equations for the half-reactions which will happen at each electrode.

3. Standard reduction potentials can be used to make predictions about oxidation and reduction reactions. These are measured relative to the standard hydrogen half-cell which is assigned an E value of 0.
- Under what conditions are the standard reduction potentials measured?
 - Where are the strongest oxidising and reducing agents found on the table?
 - The electrode in a standard hydrogen half-cell will act as either an anode or a cathode depending on the other half-cell it is reacting with. Write an equation for the standard hydrogen half-cell acting as an:
 - Anode
 - Cathode
 - Given that either hydrogen ions, **or** hydrogen gas will react at the hydrogen half-cell electrode, why is it necessary to have both present at standard conditions even though only one will react?

4. A group of students wish to determine the standard reduction potential of thulium using a galvanic cell. They make a half-cell using a strip of thulium metal and a solution of $1.00 \text{ mol.L}^{-1} \text{ Tm}(\text{NO}_3)_3$. Instead of setting up a standard hydrogen cell, they decide to create a half-cell with copper metal and copper sulfate solution with a concentration of 1.00 mol.L^{-1} . They measure a voltage of 2.66 V .
- Write the equations which would occur at the anode, cathode and overall.
 - Draw a diagram of the experiment.
 - Determine the standard reduction potential of the thulium.
 - One member of the group suggested using sodium carbonate as an electrolyte for the salt bridge. Explain whether this is a suitable suggestion.

5. Tom and Harry set up a cell with a silver electrode in a silver nitrate solution and a nickel electrode in a nickel nitrate solution. They use ammonium nitrate in the salt bridge.
- Write the equations for the half-reactions and an overall equation. Include state symbols.
 - Draw a fully labelled diagram of the cell.
 - Determine the voltage they should expect.
 - Write what would be observed in each half-cell.
 - Tom suggests they try swapping the electrodes for graphite electrodes which are inert and will not take part in the reaction. They swap the silver for a graphite one and find that the reaction proceeds as normal. They then try swapping the nickel electrode with a graphite one and find that the voltage drops to 0. Explain these observations.

6. A half-cell is set up with chlorine gas being bubbled over a platinum electrode in a solution of sodium chloride. The other half-cell contains a solution of iron(II) chloride. The other half-cell contains a solution of iron(II) chloride.



- Write equations for the process which will be occurring at the anode and the cathode and an overall redox reaction, including states.
- Determine the voltage of the cell.
- What will be observed at the iron electrode?
- How would the voltage change if fluorine gas was used in place of the chlorine gas?
- Explain what would happen if the salt bridge was removed.